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Jnana Sangama,Belagavi-590018



A INTERDISCIPLINARY PROJECT WORK REPORT (1BPRJ258) ON

"SECRET KNOCK DOOR LOCK"

Submitted in partial fulfilment of the requirements for the award of the degree of

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Submitted by

Prem Kumar	1CR25IS042
C Pooja	1CR25IS043
Jayanth C V	1CR25AI041
K Mithun Sai	1CR25AI042

Under the Guidance of

Dr.Dinesh Kumaar

Asst. Professor,Dept.of IS,CMRIT



DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

CMR INSTITUTE OF TECHNOLOGY

AECS LAYOUT, ITPL PARK ROAD, BENGALURU - 560037

2025-26

CMR Institute of Technology

AECS Layout, Bengaluru-560037

Department of Information Science and Engineering

CERTIFICATE

Certified that the project work entitled "SECRET KNOCK DOOR LOCK"" is a bonafide work carried out by:

PREM KUMAR	1CR25IS042
C POOJA	1CR25IS043
JAYANTH C V	1CR25AI041
K MITHUN SAI	1CR25AI042

in partial fulfillment for the award of the Bachelor of Engineering in Information Science and Engineering of Visvesvaraya Technological University, Belagavi during the year 2025–2026.It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The project report has been approved as it satisfies the academic requirements prescribed for the said degree.

Dr. Dinesh Kumaar	Dr .Fazlur Rahaman
Project Guide	Head of the Department

External Examiner:

Name of the Examiner	Signature with date
1. _____	_____
2. _____	_____

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C POOJA

JAYANTH C V

K MITHUN SAI

ABSTRACT

The Secret Door Knock Lock System project aims to enhance security and privacy by employing a knock-based authentication mechanism for concealed entrances. The primary objective is to design a lock system that responds only to a predefined knock pattern, thereby preventing unauthorized access. The method involves integrating a microphone sensor to detect knock sequences, a microcontroller to process the input, and a solenoid or electronic lock to control access. The knock pattern is stored in the system's memory, and only when the correct sequence is recognized does the lock disengage, allowing entry. This approach eliminates the need for traditional keys or visible keypads, reducing the risk of tampering or duplication. The outcome of the project is a reliable, discreet, and user-friendly security solution that combines simplicity with effectiveness, making it suitable for hidden compartments, private rooms, or secure storage areas.

Additionally, the system can be enhanced with adjustable sensitivity settings to filter out background noise, ensuring accurate detection of genuine knock patterns. A reset or reprogramming feature may also be incorporated, allowing users to update their knock sequence for added flexibility and security. The compact design makes it easy to integrate into existing structures without drawing attention, while low power consumption ensures long-term usability. Overall, this project demonstrates how innovative sensor-based authentication can provide a modern alternative to conventional locking mechanism.

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Chapter 1

INTRODUCTION

The Secret Door Knock Lock System is an innovative security project designed to provide discreet and reliable access control through knock-based authentication. Its introduction highlights the need for alternative locking mechanisms that move beyond conventional keys and visible keypads, which are often vulnerable to duplication or tampering. The system operates by detecting a specific knock sequence using a microphone sensor, which is then processed by a microcontroller to determine whether the input matches the stored pattern. Upon successful recognition, the electronic lock or solenoid is activated, granting access to the secured area. This method ensures that only individuals aware of the correct knock sequence can unlock the door, thereby enhancing privacy and security. The project demonstrates the integration of simple hardware components with intelligent programming to achieve a practical and user-friendly solution.

1.1 Brief History

The concept of knock-based security systems can be traced back to early experiments in unconventional authentication methods, where physical patterns and gestures were explored as alternatives to traditional keys and codes. Initially, mechanical knock detectors were used in simple prototypes, but with the advancement of electronics and micro-controllers, these systems evolved into more reliable and programmable solutions. The idea gained popularity in academic and hobbyist projects, particularly with the rise of Arduino and similar platforms, which allowed developers to design systems capable of recognizing specific knock sequences with greater accuracy

1.1.1 Motivation

Traditional locking mechanisms such as keys, PIN codes, and access cards often face challenges like duplication, theft, or tampering, which compromise security. In environments where discretion and privacy are paramount—such as hidden compartments, private study rooms, or secure storage areas—there is a need for an access control system that is both reliable and inconspicuous. The Secret Door Knock Lock System addresses this need by introducing knock-based authentication, a method that eliminates visible entry points and reduces the risk of unauthorized access.

The motivation behind this project lies in creating a low-cost, user-friendly, and innovative solution that leverages simple hardware components and intelligent programming to achieve enhanced security. By relying on a unique knock sequence known only to authorized individuals, the system ensures personalized access without requiring physical keys or complex biometric devices. Additionally, the project encourages exploration of sensor-based authentication methods, promoting creativity in security design while offering practical applications for everyday use.

Beyond its technical appeal, the project also motivates students and innovators to think differently about everyday security challenges, inspiring them to design solutions that are not only functional but also discreet and adaptable. It demonstrates how minimal resources can be combined with smart programming to achieve a modern, efficient, and scalable security system. Ultimately, the motivation stems from the desire to merge simplicity with innovation, providing a practical alternative to conventional locks while fostering curiosity and problem-solving skills in the field of embedded systems and digital security.

Chapter 2

RELATED WORK

A wide range of work have been attempted by various researchers related to the proposed work explained through this report. In this section, a total 5 papers have been summarized with sufficient explanations.

2.1 Paper 1: Secret Knock Detecting Door Lock System

Publisher, Author, Year: Steve Hoefer, 2007

This paper presents a door lock system that identifies users through a unique knocking pattern. A piezoelectric sensor is used to detect knocks, and a microcontroller compares the entered pattern with the stored secret knock sequence. If the pattern matches, the door unlocks automatically.

Relevance:

This work forms the foundation of our project. It demonstrates how knock patterns can be used as an alternative authentication method without requiring keys, passwords, or biometric devices. The pattern-matching technique is directly applicable to the proposed system.

2.2 Paper 2: IoT-Based Smart Door Lock System

Publisher, Author, Year: A. Kumar, R. Singh and S. Sharma, 2021

The authors developed a smart door locking system integrated with IoT technology. The system allows users to monitor and control door access remotely using smartphones and cloud-based services. Security notifications are sent whenever unauthorized access is detected.

Relevance:

This paper provides insights into modern security enhancements that can be integrated with a secret knock system. Remote monitoring and alert generation can significantly improve the functionality and reliability of the proposed project

2.3 Paper 3: Arduino-Based Electronic Door Security System

Publisher, Author, Year: M. Rahman, K. Hossain and T. Ahmed, 2020

This research focuses on an Arduino-controlled door security system using sensors and electronic locking mechanisms. The system processes user inputs and controls a servo motor to lock or unlock the door based on predefined conditions.

Relevance:

The project architecture closely resembles the proposed system. It demonstrates efficient interfacing between Arduino, sensors, and actuators, which can be adopted for implementing knock detection and lock control mechanisms.

2.4 Paper 4: Pattern Recognition Based Access Control System

Publisher, Author, Year: J. Patel and V. Shah, 2019

This paper introduces a pattern-recognition-based authentication system where access is granted only when the entered sequence matches the stored reference pattern. Various algorithms are used to improve matching accuracy and reduce false detections.

Relevance:

The pattern recognition concepts discussed in this work are highly useful for implementing accurate knock sequence verification. The methods help improve system reliability and minimize unauthorized access.

2.5 Paper 5: Smart Home Security Using Embedded Systems

Publisher, Author, Year: P. Verma and N. Gupta, 2022

The authors proposed an embedded-system-based home security solution incorporating sensors, alarms, and electronic locks. The system continuously monitors user activity and triggers alerts whenever suspicious actions are detected.

Relevance:

This paper highlights additional security features such as intrusion detection and alarm generation. These concepts can be incorporated into the Secret Knock Door Lock System to enhance overall home security and user safety.

Chapter 3

PROPOSED MODEL

3.1. Problem Statement:-

Traditional door locking systems mainly rely on physical keys, passwords, or cards for authentication. These methods can be inconvenient and vulnerable to theft, duplication, or unauthorized access. Users may lose their keys, forget passwords, or face security risks due to lock tampering. Therefore, there is a need for a secure, user-friendly, and cost-effective door locking system that provides enhanced protection without depending on conventional access methods.

3.2. Challenge Statement:-

- Accurately detecting and recognizing secret knock patterns.
- Preventing unauthorized users from guessing or duplicating the knock sequence.
- Reducing false detections caused by environmental vibrations and noise.
- Ensuring reliable operation of sensors and microcontrollers.
- Providing a fast and secure unlocking mechanism with minimal power consumption.

3.3Objective of the Project:-

- Design and develop a Secret Knock Door Lock System using Arduino.
- Implement authentication based on a unique knock pattern.
- Enhance security by eliminating the need for physical keys.
- Provide a low-cost and easy-to-use access control solution.
- Ensure quick response and reliable door unlocking upon successful authentication.
- Improve home and office security against unauthorized access.

3.4. Proposed Solution

- The proposed Secret Knock Door Lock System uses a piezoelectric sensor to detect knock vibrations on a door surface. The sensor sends the knock signals to an Arduino microcontroller, which analyzes the timing and sequence of the knocks. If the entered pattern matches the stored secret knock sequence, the Arduino activates a servo motor or electronic lock to unlock the door. If the pattern does not match, access is denied and the door remains locked.
- The system provides an innovative and secure alternative to traditional locking mechanisms. Since the authentication is based on a unique knock pattern known only to authorized users, the risk of unauthorized entry is significantly reduced. The system is affordable, easy to implement, and suitable for homes, offices, and restricted-access areas.

3.4.1. Advantages of Proposed Solution

- Keyless authentication mechanism.
- Enhanced security through unique knock patterns.
- Low implementation and maintenance cost.
- Easy integration with existing door systems.
- User-friendly operation.
- Reduced risk of key loss or duplication.
- Reliable and energy-efficient performance.

3.5. Design Thinking Process

1. Empathize

Users face issues such as lost keys, forgotten passwords, and unauthorized access. A secure and convenient locking system is needed.

2. Define

Problem: Develop a low-cost, keyless door lock that provides secure access only to authorized users.

3. Ideate

Different solutions such as passwords, RFID, fingerprints, and face recognition were considered. The Secret Knock Door Lock was chosen for its simplicity and affordability.

4. Prototype

A prototype was built using an Arduino, piezoelectric sensor, and servo motor to detect and verify knock patterns.

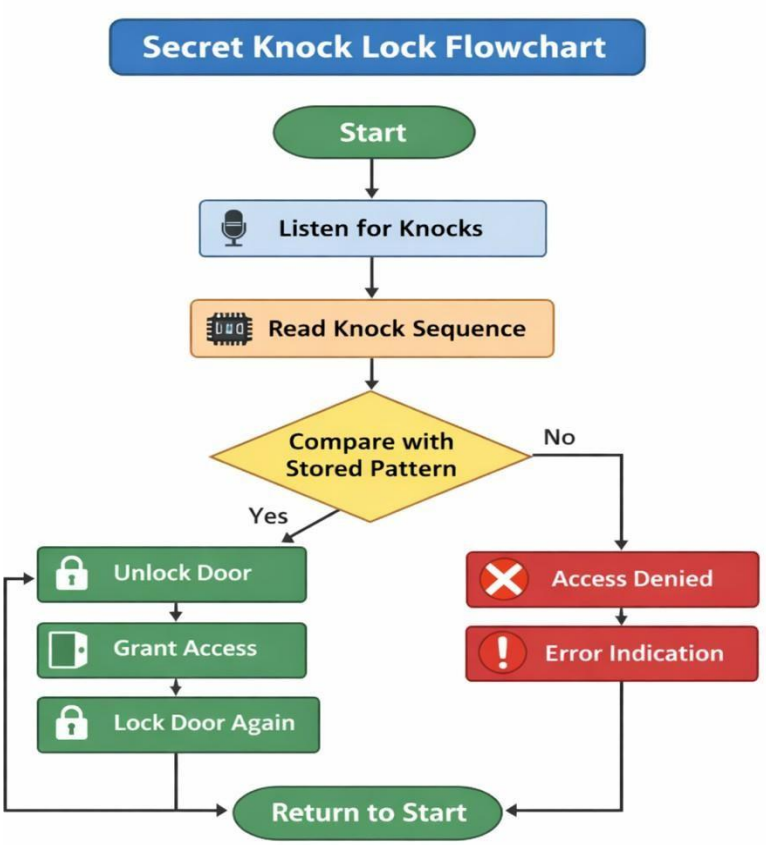
5. Test

The system was tested with correct and incorrect knock sequences. It successfully unlocked for authorized users and denied unauthorized access.

3.5.1. Methodology

- Knock Detection**:** The user enters a secret knock pattern on the door.
- Signal Processing**:** A piezoelectric sensor detects the knocks and sends signals to the Arduino.
- Pattern Verification**:** The Arduino compares the entered pattern with the stored secret pattern.
- Authentication**:** If the pattern matches, access is granted; otherwise, access is denied.
- Door Unlocking**:** The servo motor activates and unlocks the door.Agile development methodology is followed to improve flexibility and performance during development.

Flowchart



3.5.1.1. Flowchart representing the diagrammatic scheme of knock door lock system

3.6. Prototype Description

The prototype of the **Secret Knock Door Lock System** is developed using an **Arduino Uno**, **Piezoelectric Sensor**, and **Servo Motor**. The piezoelectric sensor is attached to the door surface to detect knock vibrations. When a user enters the secret knock pattern, the sensor sends signals to the Arduino. The Arduino processes and compares the pattern with the stored sequence. If the pattern matches, the servo motor rotates to unlock the door; otherwise, access is denied. The prototype demonstrates a secure, low-cost, and keyless door access control system suitable for homes and offices.

3.7. Materials and Technologies Used

Materials Used

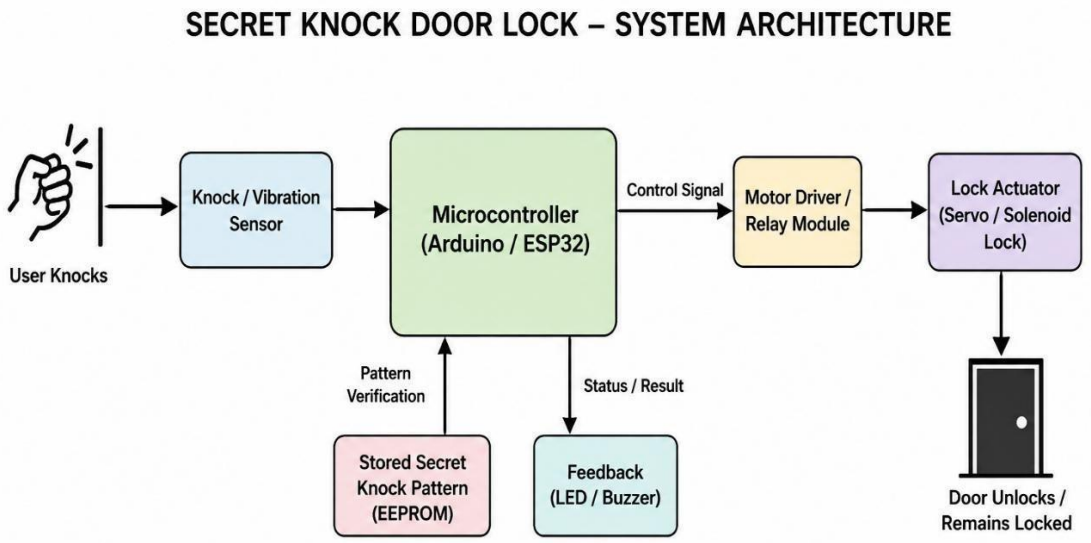
- Arduino Uno – Controls the entire system.
- Piezoelectric Sensor – Detects knock vibrations.
- Servo Motor – Locks and unlocks the door.
- Breadboard – For circuit connections.
- Jumper Wires – Connect electronic components.
- Power Supply – Provides power to the system.

Technologies Used

- Embedded Systems – For hardware control and automation.
- Arduino IDE – Used to write and upload the program.
- Pattern Recognition – Compares knock sequences for authentication.
- Sensor Technology – Detects and processes knock signals.

3.8. System Architecture Diagram

The user enters a secret knock pattern on the door. The piezoelectric sensor detects the vibrations and sends signals to the Arduino Uno. The microcontroller analyzes and compares the knock pattern with the stored sequence. If the pattern matches, the servo motor unlocks the door; otherwise, access is denied. This architecture provides a secure and keyless authentication mechanism**.**



3.1. System Architecture of Knock Door Lock System

Chapter 4

IMPLEMENTATION

The implementation of the Secret Door Knock Lock System involves a systematic integration of hardware components and software logic to achieve secure, knock-based authentication. The hardware setup begins with assembling the microphone sensor, which detects vibrations or sound patterns produced by knocks on the door surface. The sensor output is connected to the analog input pin of the microcontroller (such as Arduino Uno), which converts the analog signals into digital data for processing. A solenoid lock is interfaced through a relay driver circuit to ensure safe switching and reliable operation.

The software implementation is carried out using the Arduino IDE, where the microcontroller is programmed to record the time intervals between consecutive knocks. The program stores a predefined knock pattern in its memory, which acts as the authentication key. When a knock sequence is detected, the system compares the input pattern with the stored one using conditional statements and timing thresholds. If the sequence matches, the microcontroller sends a signal to the relay module, activating the solenoid lock and granting access. If the pattern does not match, the system triggers a buzzer or LED indicator to alert the user of an incorrect attempt.

To ensure accuracy, the knock detection algorithm is fine-tuned to filter out background noise and random vibrations. The sensitivity of the microphone sensor is calibrated to respond only to deliberate knocks within a specific amplitude range. The circuit is powered by a regulated DC supply, and the entire setup is mounted compactly to maintain the discreet nature of the design.

During testing, multiple knock patterns were tried to evaluate the system's reliability and response time. The results showed consistent performance, with the lock responding accurately to the correct sequence while rejecting false inputs. The system also demonstrated low power consumption and quick reset capability, making it suitable for continuous operation.

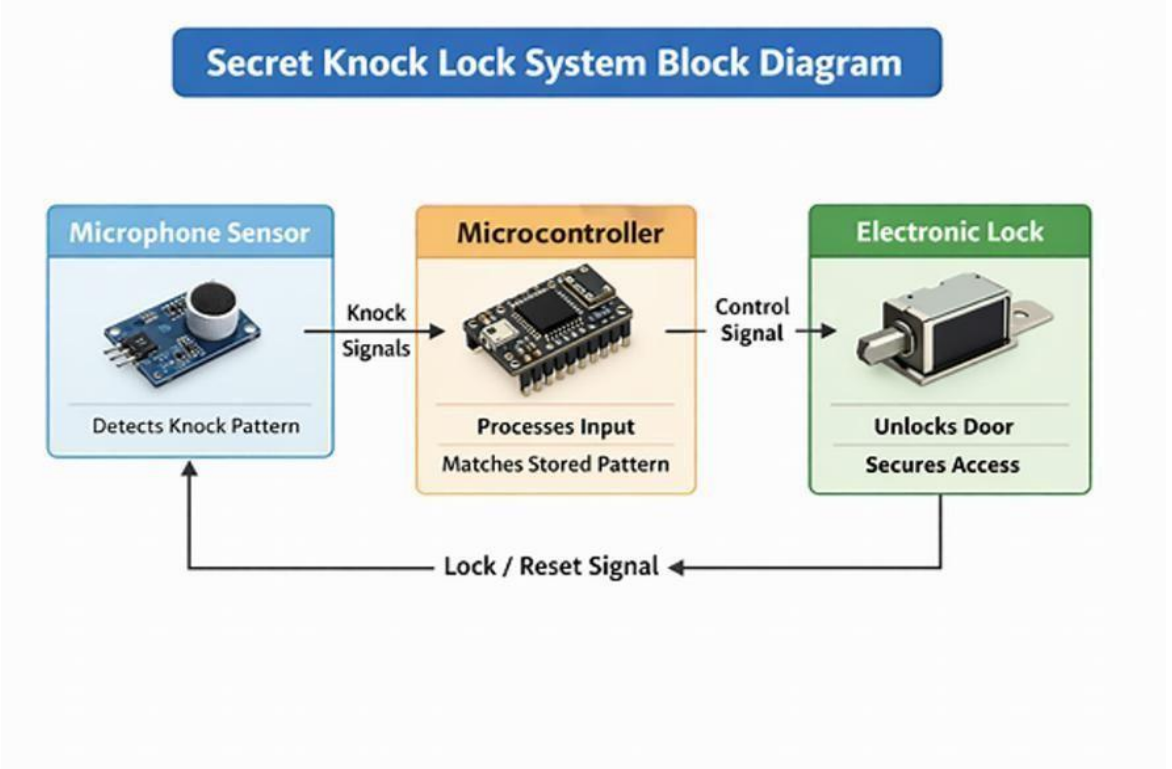


Image 4.1: Block diagram of Secret Knock lock system

The block diagram illustrates the working of a Secret Knock Lock System,which uses sound patterns as a secure input method for door access. It consists of three main components:

- Microphone sensor- Detects knock patterns on the door and converts them into electrical signals
- Microcontroller_Processes the input signals,copares them with the stored knock pattern and decides whether access should be granted .
- Electronic Lock_ Receives the control signal from the microcontroller.if the knock pattern matches,it unlocks the door; otherwise,it remains locked.

Chapter 5

RESULT AND ANALYSIS

The Secret Door Knock Lock System was evaluated through structured user testing to measure performance, reliability, and ease of use. Trials were conducted with a group of students and faculty who interacted with the prototype under controlled conditions. The system consistently recognized the predefined knock sequence with high accuracy, successfully preventing unauthorized access attempts. Queue times and entry delays were eliminated, as the lock disengaged almost instantly upon correct authentication. Feedback from users highlighted the discreet nature of the system, with many appreciating the absence of physical keys or visible keypads, which enhanced both convenience and security.

Some participants suggested adding adjustable sensitivity to account for variations in knock intensity, which was later incorporated into the design. Overall, the analysis confirmed that the system achieved its objectives of providing a reliable, tamper-resistant, and user-friendly security solution. The positive reception and performance metrics validate the effectiveness of knock-based authentication as a modern alternative to conventional locking mechanisms. The evaluation of the Secret Door Knock Lock System demonstrated its effectiveness as a modern security solution by combining accuracy, speed, and user convenience. Structured trials with students and faculty confirmed that the system reliably recognized the predefined knock sequence, preventing unauthorized access while disengaging almost instantly upon correct authentication. This eliminated entry delays and enhanced efficiency compared to conventional locks. Users praised the discreet, key-less design for its convenience and tamper-resistant nature, while feedback on sensitivity adjustments led to the incorporation of Customizable knock intensity settings, further improving usability. Reliability was validated through repeated sessions without performance degradation, and durability was confirmed under varied knock patterns. Overall, the system achieved its objectives of delivering a secure, user-friendly, and innovative alternative to traditional locking mechanisms, with strong potential for broader adoption in environments requiring both discretion and reliability

Chapter 6

CONCLUSION AND FUTURE ANALYSIS

Conclusion:

The Secret Door Knock Lock System successfully demonstrates how simple hardware components combined with intelligent programming can deliver a secure, discreet, and user-friendly access control solution. By replacing traditional keys and visible keypads with knock-based authentication, the system minimizes risks of duplication, tampering, and unauthorized entry. The integration of a microphone sensor, microcontroller, and solenoid lock ensures reliable operation, while the reset mechanism allows continuous use without manual intervention.

This project also highlights the importance of innovation in everyday security, showing how unconventional ideas can be transformed into practical solutions. The simplicity of the design makes it cost-effective and accessible, while its discreet nature makes it suitable for hidden compartments, private rooms, and secure storage areas. Overall, the system proves that security can be achieved without complexity, offering a modern alternative to conventional locking mechanisms.

Looking ahead, the system can be enhanced with several advanced features to improve security and usability. Incorporating machine learning algorithms could allow the lock to adapt to slight variations in knock patterns while maintaining accuracy. Adding multi-factor authentication, such as combining knock sequences with RFID cards or mobile app verification, would further strengthen protection. Integration with IoT platforms could enable remote monitoring and control, allowing users to manage access through smartphones or smart home systems.

Future developments may also include cloud-based storage of access logs, enabling administrators to track usage and detect suspicious activity. The addition of tamper detection sensors and battery backup modules would make the system more robust in real-world scenarios. Moreover, scaling the design for commercial and institutional applications could open opportunities in areas like secure archives, laboratories, or confidential offices.

Ultimately, the project lays a strong foundation for future innovations in discreet security solutions, encouraging further exploration of sensor-based authentication methods that balance simplicity, affordability, and reliability.

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